UNIT-3: Vector Calculus
Appearance Year: 2025-26

1. Gradient, Divergence, Curl:
   - Show that gradient field describing a motion is irrotational.
   - Find directional derivative of divergence of F = xy i + xy^2 j + z^2 k at (2,1,2) normal to sphere x^2+y^2+z^2=9.
2. Vector Integral Theorems:
   - Apply Green's theorem for int_c [(y - sin x)dx + cos x dy] over triangle y=0, x=pi/2, y=(2/pi)x.
   - Apply Gauss's divergence theorem for F = (2x^2-3z)i - 2xy j - 4x k over plane x+2y+z=4.